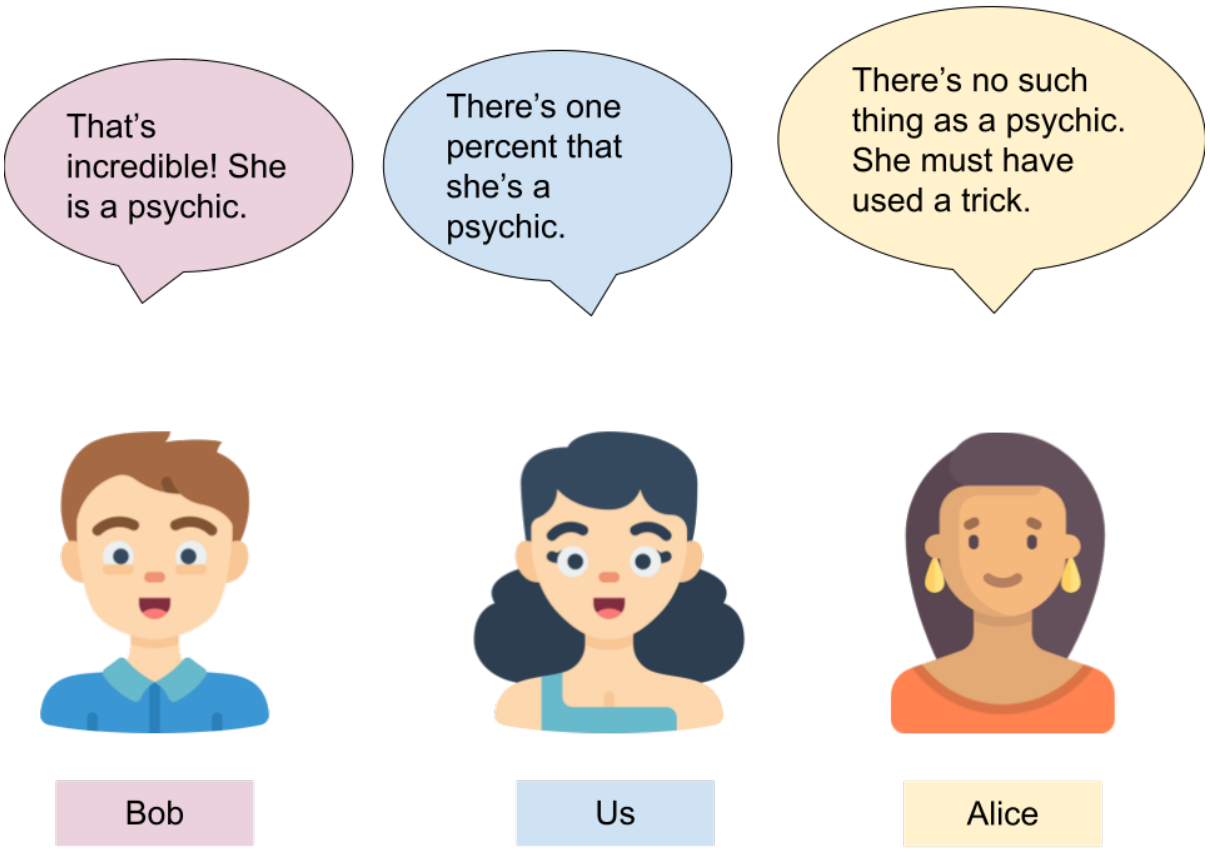


Figure 3.1 Reaction among our group of friends after seeing a person correctly guesses the secret answer one hundred times. Each person reached a different conclusion due to their prior belief.

How can all three of us observe the same event (the person at the carnival guessing correctly one hundred times) and arrive at different conclusions? To answer this question, consider the process of updating one's belief using Bayes' theorem.

SIDEBAR **More reading on Bayes' theorem**

We won't go through the exact math here, but I recommend Chapters 1 and 2 of Will Kurt's Bayesian Statistics the Fun Way if you want to see how this is exactly done. We will only sketch this out on a high level.

1. Each person starts out with a specific prior probability that the woman is a psychic.
2. Then, we observe the event that she guesses our number correctly once.
3. We then compute the likelihood terms. First, the likelihood that her guess is correct given that she's indeed a psychic is exactly 1, since a true psychic can always pass this test. Second, the likelihood that her guess is correct, given that she's *not* a psychic, is 1 out of 10. This is because each time we are randomly choosing a number between 0 and 9, so any guess among these ten options has an equal chance of being correct: 1 out of 10.
4. Finally, we update our belief by computing posterior probability that this person is not a psychic, by combining our prior with these likelihood terms. Specifically, this posterior will be proportional to the prior and the first likelihood term *multiplied* together.

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